

$$\text{Ratio} = 0.020$$

**28 C**  $p = mv = (9.11 \times 10^{-31})(1.50 \times 10^6) = 1.37 \times 10^{-24} \text{ kg m s}^{-1}$   
 $\Delta p = 0.2\% \times p = 0.002 p = 0.002(1.37 \times 10^{-24}) = 2.73 \times 10^{-27} \text{ kg m s}^{-1}$   
 $\Delta x = h/(\Delta p) = 6.63 \times 10^{-34} / (2.73 \times 10^{-27}) = 2.4 \times 10^{-7} \text{ m}$

**29 A** Energy released = BE of products – BE of reactants